



Northern University

Of Business and Technology Khulna

LAB Report - 01

Course Code: CSE 1202

Course Title: Object Oriented Programming Lab

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Experiment No: 01**Experiment Name:**

Program to Calculate the Circumference of a Circle Using C++

Introduction

A circle is a basic geometric shape defined as the set of all points that are equidistant from a fixed point known as the center. The distance from the center to any point on the circle is called the radius.

The circumference of a circle represents the total distance around its boundary. It is calculated using the mathematical formula:

$$C = 2\pi r$$

Where:

- C is the circumference
- r is the radius
- π is a constant approximately equal to 3.1416

In this experiment, a C++ program is developed to calculate the circumference of a circle by taking the radius as input from the user. The program uses fundamental programming concepts such as variables, constants, input/output operations, and arithmetic expressions. The value of π is declared as a constant to ensure accuracy and maintain good coding practice.

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    double radius, circumference;
```

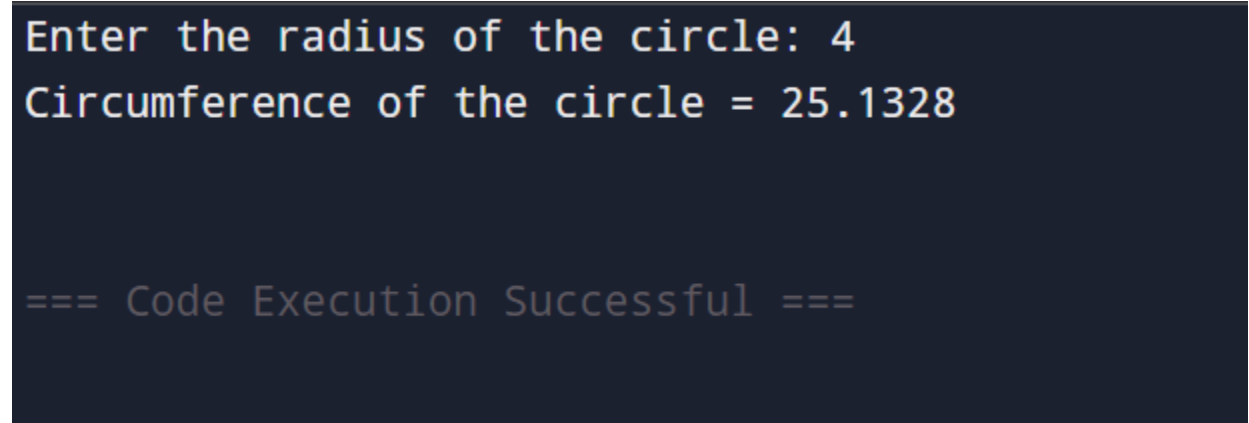
```
    const double PI = 3.1416;
```

```
    cout << "Enter the radius of the circle: ";
```

```
    cin >> radius;
```

```
    circumference = 2 * PI * radius;
```

```
cout << "Circumference of the circle = " << circumference << endl;  
  
return 0;  
}
```

Output:

```
Enter the radius of the circle: 4  
Circumference of the circle = 25.1328  
  
=== Code Execution Successful ===
```

Discussion and Conclusion

The experiment was successfully completed by implementing the mathematical formula $C=2\pi r$ in C++. The program efficiently accepts user input, performs the required computation, and displays the accurate result.

Proper programming practices were followed, including the use of the double data type for precision and declaring π as a constant to maintain code reliability. The structure of the program is clear, logical, and easy to understand. Since it performs a fixed number of operations, the time complexity is constant $O(1)$.

In conclusion, the objective of calculating the circumference of a circle using C++ was achieved successfully. This experiment effectively demonstrates the application of mathematical concepts in programming and strengthens fundamental coding skills.